

VENNAPUSA DEVENDRA REDDY

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ABOUT

AI Engineer with hands-on experience in **Generative AI, LLMs, AI Agents, RAG, and semantic search**. Experienced in building AI applications using **GPT-4, Azure OpenAI, Google Gemini, AWS Bedrock, LangChain, LangGraph, Hugging Face, and FAISS**. Strong expertise in **LLM-powered automation, autonomous AI agents, intelligent document processing, prompt engineering, embeddings, and AI agent orchestration**. Proficient in developing AI-powered applications using **Python, FastAPI, REST APIs, React.js, Streamlit, Docker, AWS Lambda, and Git**. “

SKILLS

Programming Languages — Python

Back End — Python (FastAPI, Flask)

Database — MySQL, PostgreSQL, SQLite, ChromaDB, FAISS

Machine Learning & Deep Learning — Machine Learning, Deep Learning, Scikit-learn, TensorFlow, PyTorch, NLP

Generative AI & LLMs — RAG, Prompt Engineering, LLM Applications, GPT-4, Azure OpenAI, Google Gemini, LLaMA, Hugging Face, Transformers, Embeddings, Semantic Search

AI Frameworks & Agentic AI — LangChain, LangGraph, AutoGen, CrewAI, AI Agents, Multi-Agent Systems, Agent Design, Tool Calling, Memory Management

Vector Databases — ChromaDB, FAISS

Cloud & AI Services — Microsoft Azure (Azure OpenAI, Azure AI Services), AWS, AWS Bedrock, AWS Lambda

Data & Integration — REST APIs, API Integration, Data Pipelines

Visualization & Deployment — Streamlit, Plotly, Chart.js, Docker, Docker Compose

Version Control & Collaboration — Git, GitHub, Bitbucket

EXPERIENCE

TeckyBot

Jan 2025 – Apr 2025

Artificial Intelligence Intern

- Developed **Generative AI applications and an AI Healthcare Chatbot** using **Google Gemini LLM, RAG, prompt engineering, and conversational AI** for intelligent healthcare assistance and medical document summarization.

Synycs Enterprises Private Limited

Mar 2026 – Apr 2026

AI Intern

- Engineered a **Dynamic Pricing Intelligence Platform** using **Machine Learning, automated data pipelines, competitor price tracking, anomaly detection, and GenAI-powered reporting** for actionable pricing insights.

PROJECTS

Atlas – Autonomous AI Code Execution Agent

Tech Stack: Python, Streamlit, FastAPI, LangChain, Lang Azure OpenAI, GPT-4, FAISS, Docker, Git, REST APIs, Prompt Engineering, JSON, GitHub

Project Description: Developed an enterprise-grade Autonomous AI Code Execution Agent that analyzes software repositories, generates structured execution plans using Large Language Models, performs automated code modifications, validates generated code, and produces comprehensive execution reports. The platform supports

local repositories, ZIP uploads, and remote Git repositories with automated verification and self-healing mechanisms.

Roles & Responsibilities

- Designed an autonomous repository analysis pipeline using **LLMs** to discover project architecture, frameworks, and entry points, generating structured execution plans for automated code modifications.
- Built an automated code execution engine with **syntax validation and self-healing mechanisms**, integrating Node.js-based verification to ensure reliable and high-quality generated code.
- Implemented flexible repository ingestion supporting **local paths, ZIP uploads, and remote Git repositories**, with automatic project root resolution and secure handling to prevent exposure of sensitive files.
- Developed a unified **Streamlit interface** for repository upload, execution monitoring, and downloadable reports; containerized the application using **Docker**.

AI-Powered Clinical Trial Eligibility Engine

Tech Stack: Python, FastAPI, React.js, Vite, SentenceTransformers, Hugging Face, FAISS, SQLAlchemy, SQLite, OCR, Pandas, Docker, REST APIs, Git

Project Description: Developed an AI-powered Clinical Trial Eligibility Engine that automates patient pre-screening by analyzing clinical trial protocols and patient records. The solution combines semantic search, vector embeddings, document processing, and rule-based validation to identify inclusion and exclusion criteria and generate explainable eligibility decisions.

Roles & Responsibilities

- Architected an AI-driven eligibility engine using **semantic search with SentenceTransformers and FAISS** and rule-based validation to match patient profiles against clinical trial criteria, including silent clinical triggers such as eGFR thresholds.
- Built document ingestion pipelines to parse **patient records and clinical trial protocols from PDF and Excel files** using OCR and data extraction techniques.
- Developed **RESTful APIs using FastAPI** and relational data models using **SQLAlchemy/SQLite** for patient management, protocol upload, and eligibility evaluation.
- Built a responsive **React.js/Vite frontend** with secure authentication and eligibility visualization, and containerized the full-stack application using **Docker Compose**.

EDUCATION

B.Tech – Computer Science and Engineering (AI & DS)

2021 – 2025

Audisankara Institute of Technology, Gudur

CGPA: 8.4